23CSE111 OBJECT ORIENTED PROGRAMMING
S2 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus
CASE STUDY
PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : 10
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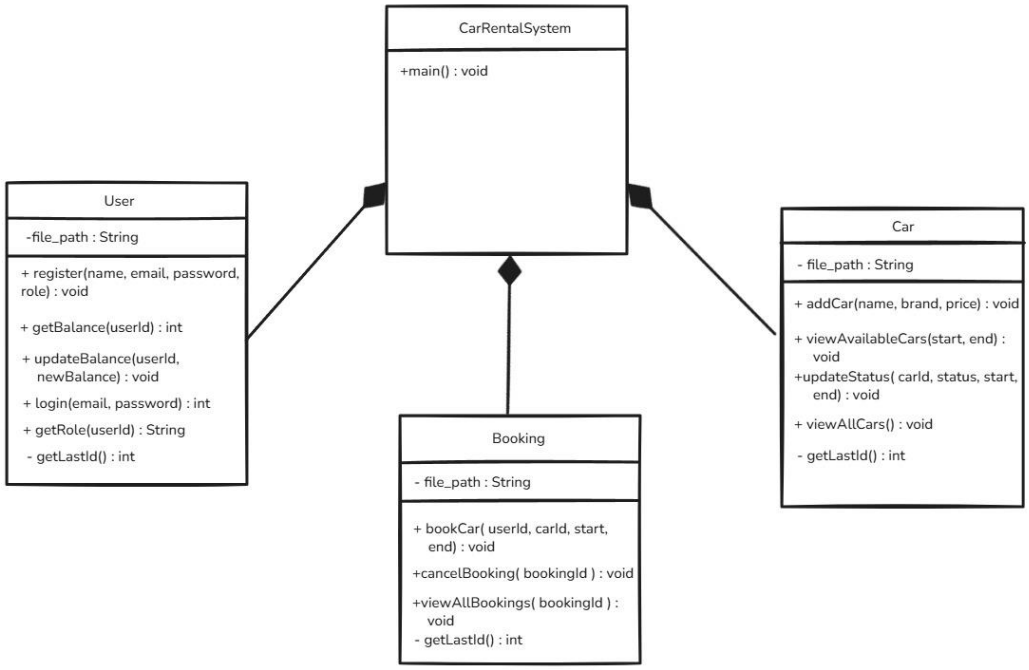
1. Problem Title : ONLINE CAR RENTAL SYSTEM

STRUCTURAL DESIGN

2. Class diagram (draw class diagram that shows different classes and their rela onships. Use UML nota ons)



BEHAVIUORAL DESIGN

3. Use-case diagram

a. List of actors

1. Admin
2. User

b. List of use-cases

1) Create a new account with name, email, password

2) Authenticate using email & password

3) View Available Cars
Enter start & end date
System shows only available cars

4) Book Car
Select car
Enter dates + km
Confirm payment
Booking created

5) View My Bookings
See list of bookings with booking ID

6) Cancel Booking
Enter booking ID

Booking marked cancelled
Car becomes available

7) View Wallet Balance
Check current balance

8) Add Balance (Top-up)
Add money to wallet

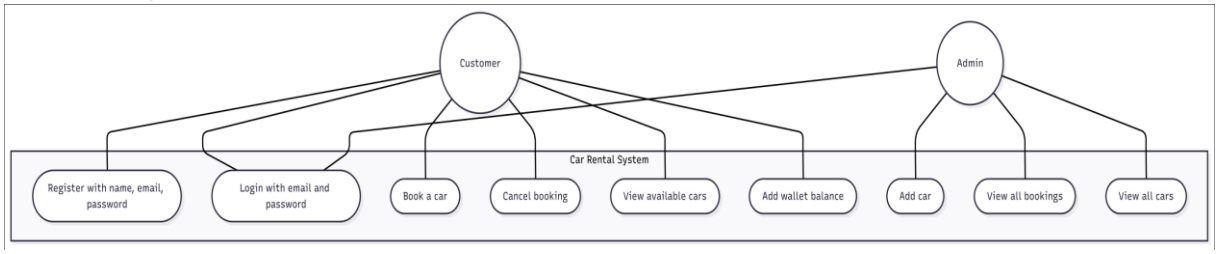
9)Login as Admin
Access admin panel

10) Add Car
Add new car details (name, brand, price)

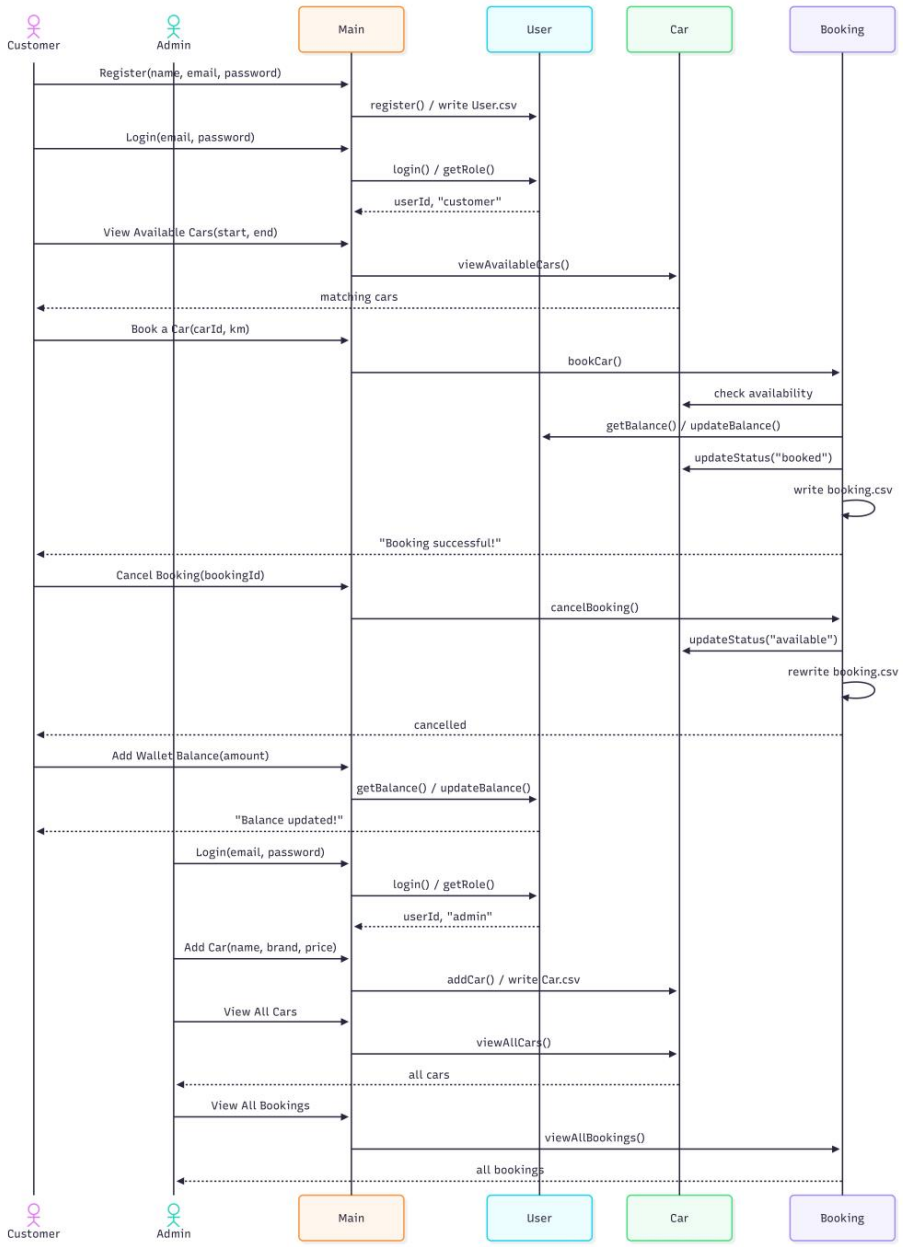
11) View All Cars
See all cars with status

12) View All Bookings
See complete booking records

c. Use case diagram (use UML notations)

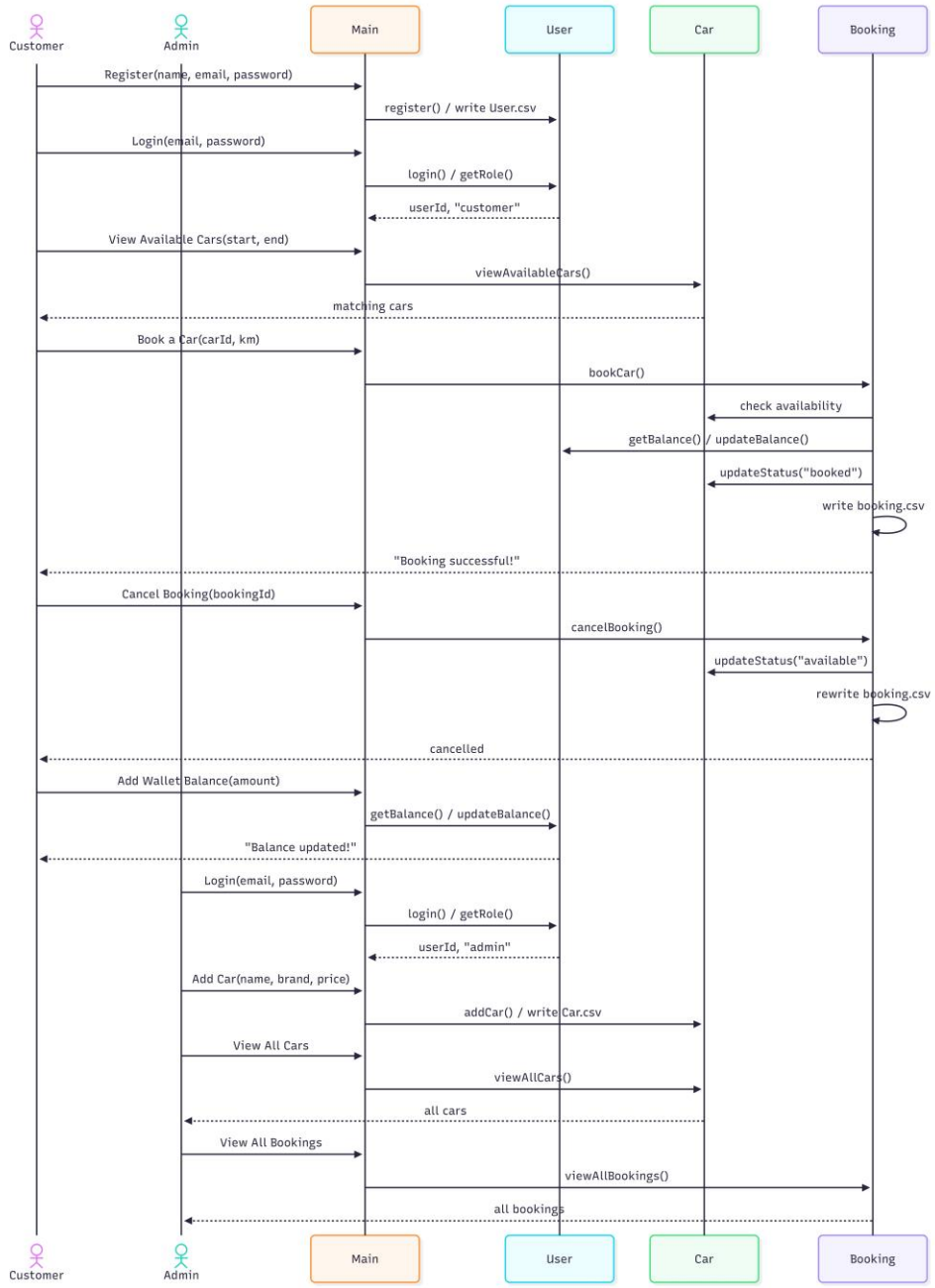


4. Sequence diagram (use UML nota ons)
   a. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.



b. Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

A new user registers by providing their personal details such as name, email, and password. The system stores this information in the csv file User.csv . Later, the user logs in using their credentials. The system verifies the details and grants access if authentication is successful.


Customer
Admin
Main
User
Car
Booking
Register(name, email, password)
register() / write User.csv
Login(email, password)
login() / getRole()
userId, "customer"
View Available Cars(start, end)
viewAvailableCars()
matching cars
Book a Car(carId, km)
bookCar()
check availability
getBalance() / updateBalance()
updateStatus("booked")
write booking.csv
"Booking successful!"
Cancel Booking(bookingId)
cancelBooking()
updateStatus("available")
rewrite booking.csv
cancelled
Add Wallet Balance(amount)
getBalance() / updateBalance()
"Balance updated!"
Login(email, password)
login() / getRole()
userId, "admin"
Add Car(name, brand, price)
addCar() / write Car.csv
View All Cars
viewAllCars()
all cars
View All Bookings
viewAllBookings()
all bookings
Customer
Admin
Main
User
Car
Booking